$= 1 - 0.0183 \times 23.667$

$= 0.567$

Example 10.12

In a Poisson distribution $3P(X= 2) = P(X = 4)$. Find its parameter $'\lambda$

Solution:

The *pmf* of Poisson distribution is $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$, $x= 0, 1, 2, ...\infty$,

Given $3\ P(X = 2)=P(X = 4)$

$$3.\frac{e^{-\lambda}\lambda^2}{2!} = \frac{e^{-\lambda}.\lambda^4}{4!}$$

$$\lambda^2 = \frac{3 \times 4!}{2!} = 36$$

$\therefore \lambda = 6$ as $\lambda > 0$

Example 10.13

Find the skewness and kurtosis of a Poisson variate with parameter 4.

Solution:

$$\lambda = 4$$

$$\text{Skewness} = \frac{1}{\sqrt{\lambda}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$\text{Kurtosis} = \frac{1}{\lambda} = \frac{1}{4}$$

Example 10.14

If there are 400 errors in a book of 1000 pages, find the probability that a randomly chosen page from the book has exactly 3 errors.

Solution:

Let X denote the number of errors in pages

$X \sim P(\lambda)$

$\therefore P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ $\quad x= 0, 1, 2, ...\infty$

The average number of errors per page $= \frac{400}{1000}$

i.e., $\lambda = \frac{400}{1000} = 0.4$